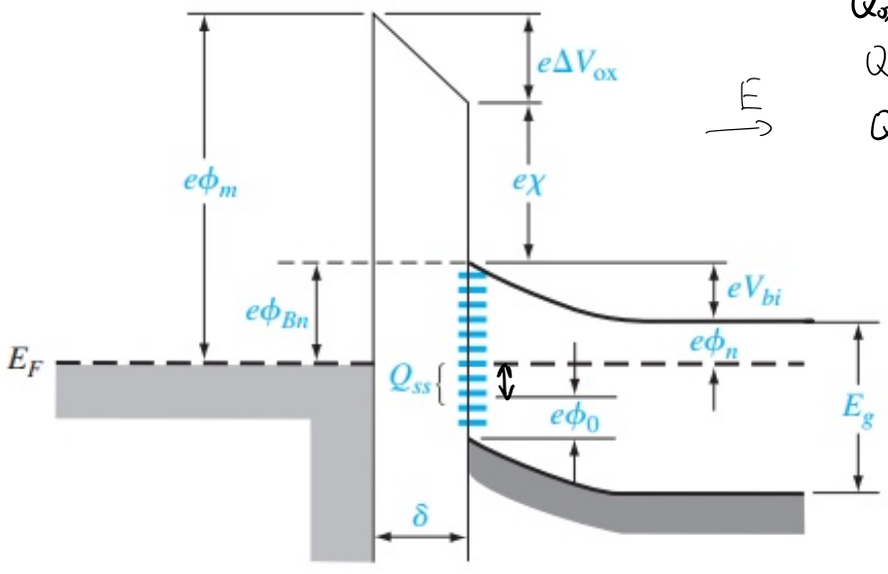
Homework 9 (due on 6-8-2026) optional

1. If there are some interface states in the metal-semiconductor junction as shown in Figure 9.6 in the book as below. (20 pts)



$Q_{ox} = C_{ox} \cdot \Delta V_{ox}$

$Q_{ox} = \frac{\epsilon_{ox}}{\delta} \cdot (\phi_m - \chi - \phi_{Bn})$

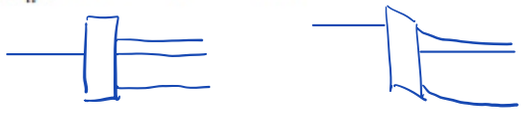
$Q_{ss} = -e \cdot D_{it}$

$= e \cdot D_{it}$

Figure 9.6 | Energy-band diagram of a metal–semiconductor junction with an interfacial layer and interface states.

Please derive the equation 9.16 in book as below.

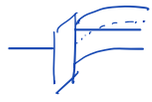
$$(E_g - e\phi_0 - e\phi_{Bn}) = \frac{1}{eD_{it}}\sqrt{2e\epsilon_s N_d(\phi_{Bn} - \phi_n)} - \frac{\epsilon_i}{eD_{it}\delta}[\phi_m - (\chi + \phi_{Bn})]$$



2. Plot the energy band diagram through the MOS structure with an n-type substrate for accumulation, depletion and inversion regions. Assuming the flat band voltage is zero for simplicity. (30 points)

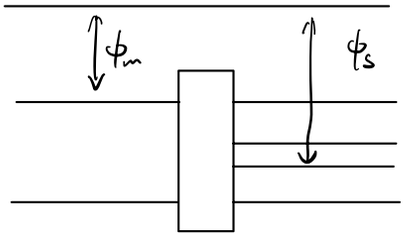


3. Derive the threshold voltage for the MOS structure with an n-type substrate. (30 points)



$\rightarrow E_F = E_c$

4. An n^+ polysilicon gate–silicon dioxide–silicon MOS capacitor has an oxide thickness of tox = 18 nm = 180 Å and a doping of Na = 1015cm^{-3}. The oxide charge density is $Q_{ss} = 6 \times 10^{10}$ cm^{-2}. Calculate the (a) flat-band voltage and (b) threshold voltage. (30points)



$V_{ox} = -\frac{Q_{ss}}{C_{ox}}$

$\phi_{ms} = -\frac{1}{2}E_g - \phi_F$, $V_{Flat} = -\frac{1}{2}E_g - \phi_F - \frac{Q_{ss}}{C_{ox}}$

1. D_{it} : surface state density (states/cm²-eV)

$$Q_{ss} = e \cdot D_{it} \cdot (E_{F_i} - E_F)$$

$$\Rightarrow Q_{ss} = -e \cdot D_{it} \cdot (E_g - e\phi_s - e\phi_{Bn})$$

$$e \cdot \Delta V_{ox} = e\phi_m - e\chi - e\phi_{Bn}$$

$$E_{ox} = \frac{\Delta V_{ox}}{\delta} = \frac{1}{\delta}(\phi_m - \chi - \phi_{Bn})$$

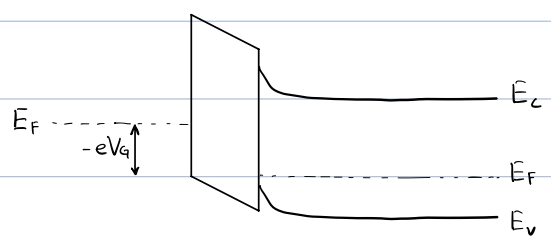
$$\begin{cases} E_{ss} = \dfrac{eN_D}{\epsilon_s} \cdot W \\ W = \sqrt{\dfrac{2V_{bi} \cdot \epsilon_s}{eN_D}} \end{cases} \Rightarrow E_{ss} = \sqrt{\frac{2V_{bi} \cdot eN_D}{\epsilon_s}} = \sqrt{\frac{2e(\phi_{Bn} - \phi_n)}{\epsilon_s}}$$

Boundary condition of electric field : $E_{ox} - E_{ss} = \dfrac{Q_{ss}}{\epsilon_s}$

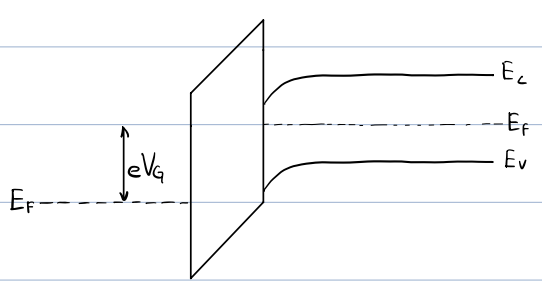
$$\Rightarrow \frac{1}{\delta}(\phi_m - \chi - \phi_{Bn}) - \sqrt{\frac{2e(\phi_{Bn} - \phi_n)}{\epsilon_s}} = -\frac{e}{\epsilon_s} D_{it} \cdot (E_g - e\phi_s - e\phi_{Bn})$$

$$\Rightarrow E_g - e\phi_s - e\phi_{Bn} = \frac{1}{eD_{it}} \sqrt{2e\epsilon_s(\phi_{Bn} - \phi_n)} - \frac{\epsilon_s}{\delta \cdot D_{it}}(\phi_m - \chi - \phi_{Bn})$$

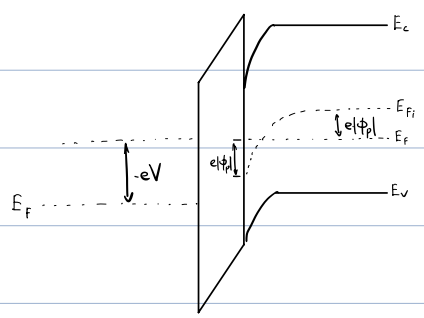
2. ① Accumulation : $V_G < 0$



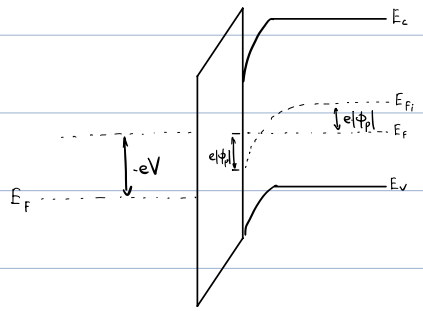
② depletion : $0 < V_G < V_{th}$



③ inversion : $V_G = V_{th} = \dfrac{Q_{ss}}{C_{ox}} + 2|\phi_p|$



3.



$V_G = V_s + V_{ox} + V_{Flat}$

$V_s = 2|\phi_p|$

$\Rightarrow V_G = 2|\phi_p| + \frac{2\sqrt{eN_A\epsilon_s|\phi_p|}}{\epsilon_{ox}} t_{ox} + V_{Flat}$

$V_s = \frac{1}{2} E_s \cdot W$

$E_s = \frac{eN_A}{\epsilon_s} \cdot W$

$V_s = \frac{eN_A}{2\epsilon_s} \cdot W^2$

$\Rightarrow W = \sqrt{\frac{2\epsilon_s V_s}{eN_A}}$

$E_{ox} = \frac{E_s \cdot \epsilon_s}{\epsilon_{ox}} = \frac{eN_A}{\epsilon_{ox}} \cdot W = \frac{\sqrt{2eN_A\epsilon_s V_s}}{\epsilon_{ox}}$

$V_{ox} = E_{ox} \cdot t_{ox} = \frac{\sqrt{2eN_A\epsilon_s V_s}}{\epsilon_{ox}} \cdot t_{ox}$

$0 < x < W$: $\rho = -eN_A$. $E = -\frac{eN_A}{\epsilon_s}$. $E(0^+) = \frac{eN_A W}{\epsilon_s}$

$-t_{ox} < x < 0$: $\rho = 0$. $E(0^-) = \frac{eN_A W}{\epsilon_{ox}}$

$$V_G = \int_{-\infty}^{+\infty} E \cdot dx = \frac{eN_A W t_{ox}}{\epsilon_{ox}} + \frac{eN_A W^2}{2\epsilon_s} + V_{Flat} = V_{ox} + V_s + V_{Flat}$$

when $V_s = 2|\phi_p|$, $V_G = V_{th}$, $W = \sqrt{\frac{4|\phi_p|\epsilon_s}{eN_A}}$

$$\Rightarrow V_{th} = \frac{eN_A W t_{ox}}{\epsilon_{ox}} + 2|\phi_p| + V_{Flat}$$

$$V_{th} = \frac{2\sqrt{eN_A|\phi_p|\epsilon_s}}{\epsilon_{ox}} \cdot t_{ox} + 2|\phi_p| + V_{Flat}$$

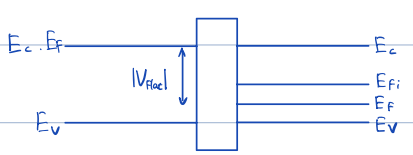
4. (a) $t_{ox} = 1.8 \times 10^{-10}\,cm$, $N_A = 10^{15}\,cm^{-3}$, $Q_{ss} = 6 \times 10^{10}\,cm^{-2}$, $E_g = 1.12\,eV$

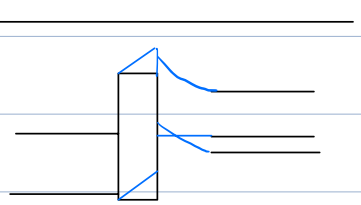
$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = 1.92 \times 10^{-7}\,F/cm$$

$$\varphi_F = \frac{kT}{e} \cdot \ln\frac{N_A}{n_i} = 0.299\,V$$

$$\Phi_{ms} = -\frac{E_g}{2e} - \varphi_F = -0.859\,V$$

$$\Rightarrow V_{Flat} = \phi_{ms} + \left(-\frac{Q_{ss}}{C_{ox}} \cdot e\right) = -0.909\,V$$





(b) $V_{th} = V_{Flat} + 2\phi_F + \frac{\sqrt{2\epsilon_s \cdot eN_A \cdot 2 \cdot |\phi_p|}}{C_{ox}} = -0.241\,V$